

10.2.8

EE25BTECH11002 - Achat Parth Kalpesh

Question:

The shortest distance from the point (2,7) to the circle $x^2 + y^2 - 14x - 10y - 151 = 0$ is equal to 5

Solution:

The general equation of a circle is

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (0.1)$$

where

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -7 \\ -5 \end{pmatrix}, \quad f = -151 \quad (0.2)$$

The center and radius are given by

$$\mathbf{C} = -\mathbf{u} = \begin{pmatrix} 7 \\ 5 \end{pmatrix} \quad (0.3)$$

$$r = \sqrt{\mathbf{u}^T \mathbf{u} - f} \quad (0.4)$$

$$r = 15 \quad (0.5)$$

Let the given point be $\mathbf{P}$

$$\mathbf{P} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad (0.6)$$

The distance between $\mathbf{P}$ and $\mathbf{C}$ is

$$\|\mathbf{P} - \mathbf{C}\| = \sqrt{(\mathbf{P} - \mathbf{C})^T (\mathbf{P} - \mathbf{C})} \quad (0.7)$$

$$= \sqrt{(2-7)^2 + (7-5)^2} \quad (0.8)$$

$$= \sqrt{25 + 4} = \sqrt{29} \quad (0.9)$$

Substituting (0.6) in (0.1)

$$\mathbf{P}^T \mathbf{P} + 2\mathbf{u}^T \mathbf{P} + f < 0 \quad (0.10)$$

The shortest distance of the point inside the circle is given as

$$d = r - \|\mathbf{P} - \mathbf{C}\| \quad (0.11)$$

Hence,

$$d = 15 - \sqrt{29} \quad (0.12)$$

Thus the shortest distance from the point (2,7) to the circle $x^2 + y^2 - 14x - 10y - 151 = 0$ is not equal to 5, it's $15 - \sqrt{29}$

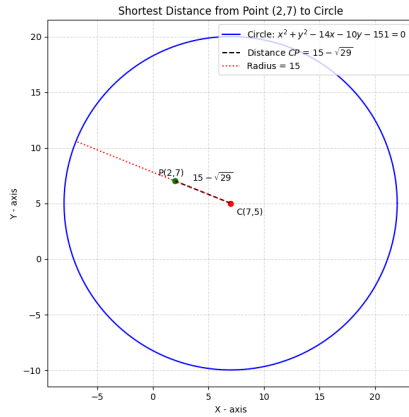


Fig. 0.1: Shortest Distance from Point (2,7) to Circle